



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS23432 – Software Construction

(REGULATION 2023)

**RAJALAKSHMI ENGINEERING COLLEGE
Thandalam, Chennai-602015**

Name:

Register No:

Year / Branch / Section:

Semester:

Academic Year:

INDEX

S.No.	Date	Title
1.		Azure Devops Environment Setup.
2.		Azure Devops Project Setup and User Story Management.
3.		Setting Up Epics, Features, And User Stories for Project Planning.
4.		Sprint Planning.
5.		Poker Estimation.
6.		Designing Class and Sequence Diagrams for Project Architecture.
7.		Designing Architectural and ER Diagrams for Project Structure.
8.		Testing – Test Plans and Test Cases.
9.		Load Testing and Pipelines.
10.		GitHub: Project Structure & Naming Conventions.

EXP NO: 1	AZURE DEVOPS ENVIRONMENT SETUP
-----------	---------------------------------------

Aim:

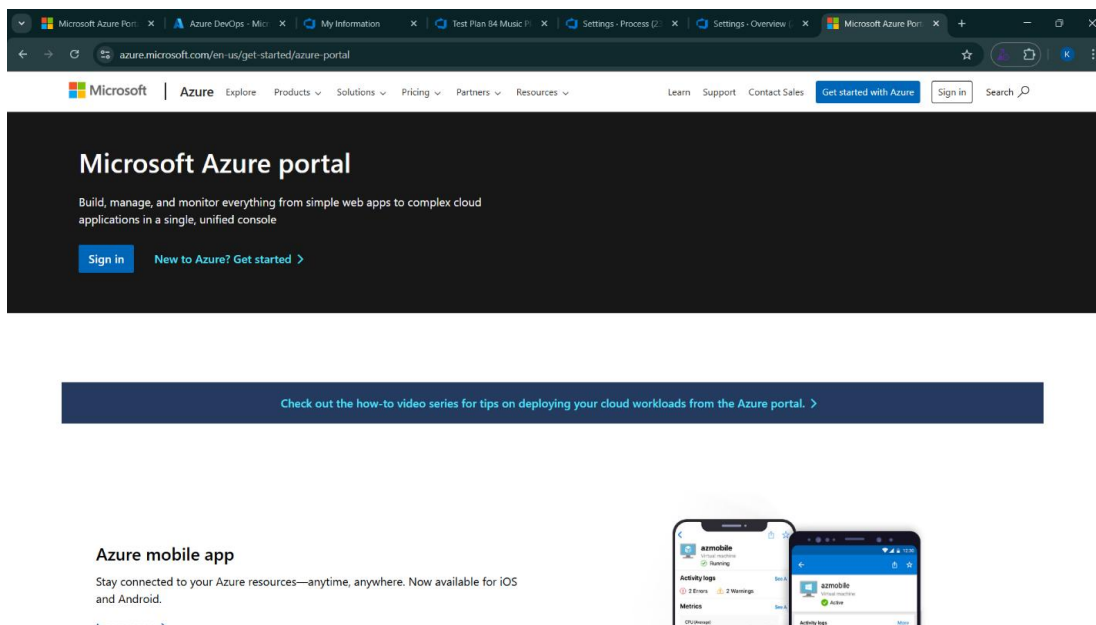
To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

INSTALLATION

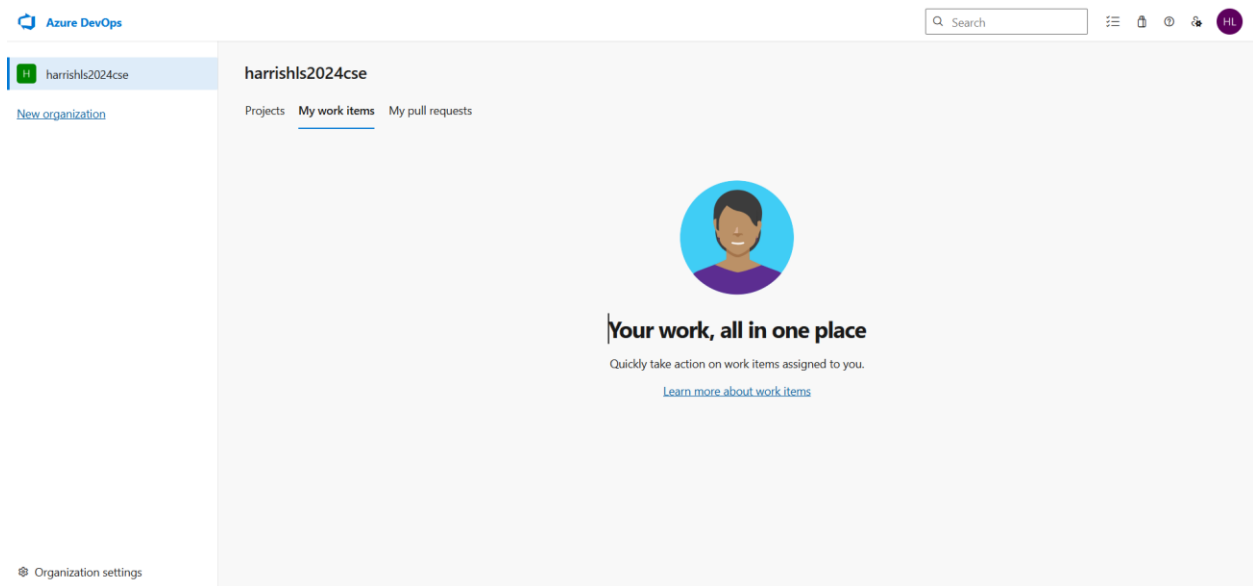
1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

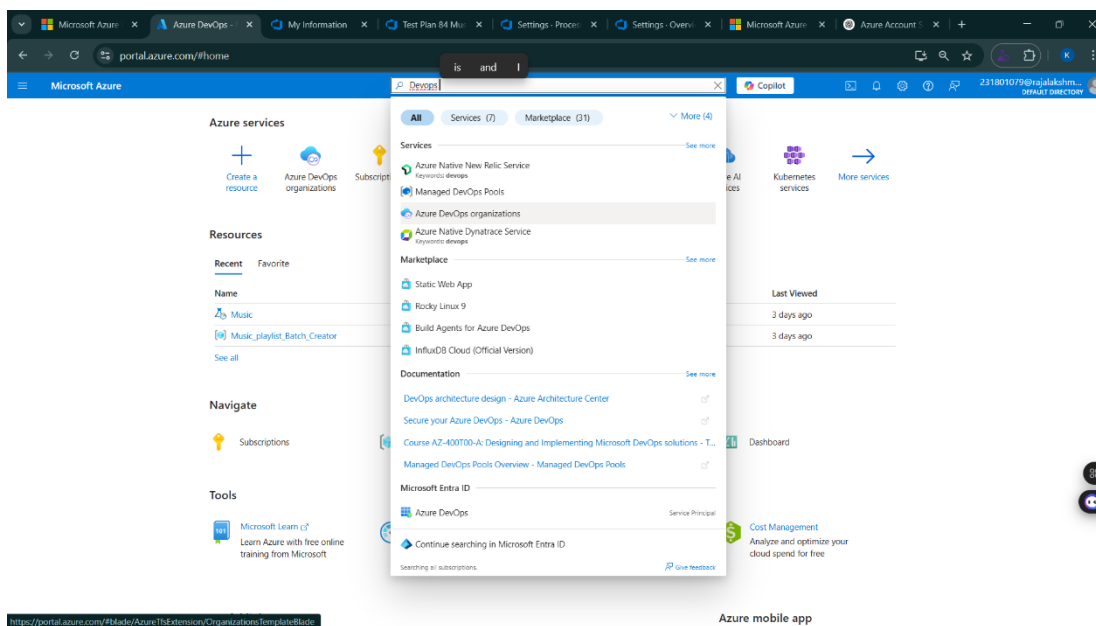
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



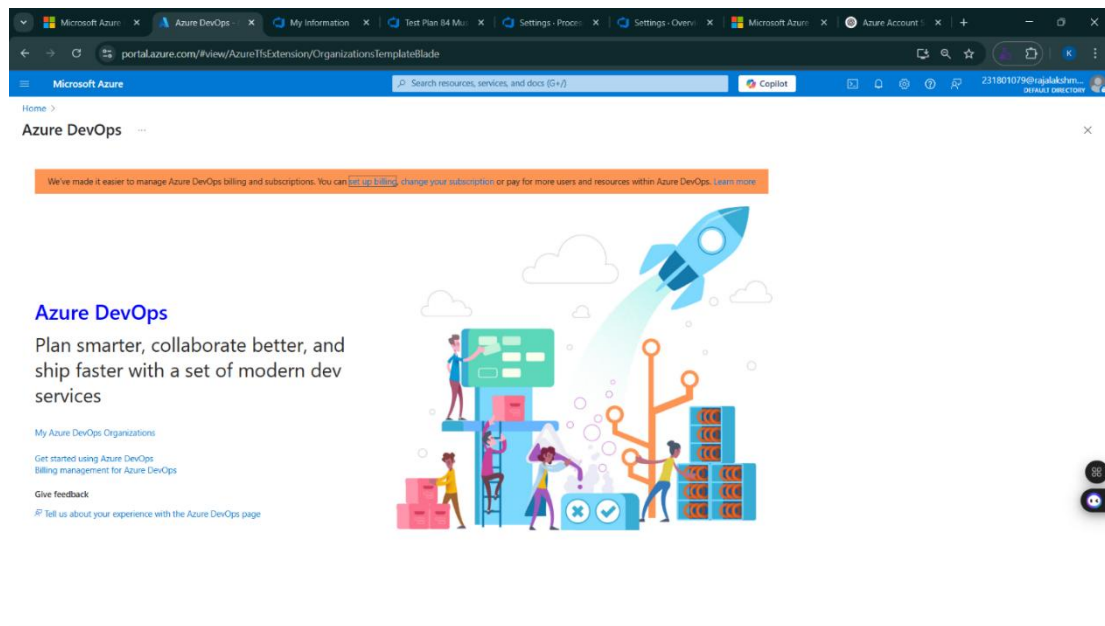
2. Azure home page



3. Open DevOps environment in the Azure platform by typing **Azure DevOps Organizations** in the search bar.



4. Click on the **My Azure DevOps Organization** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

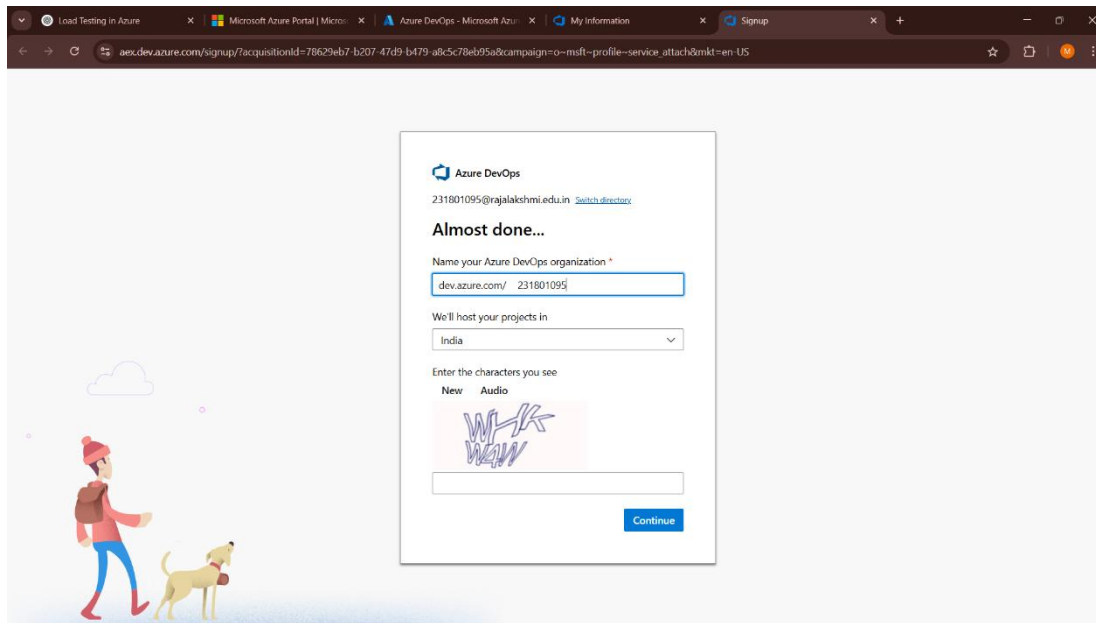
Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT
-----------	---

Aim:

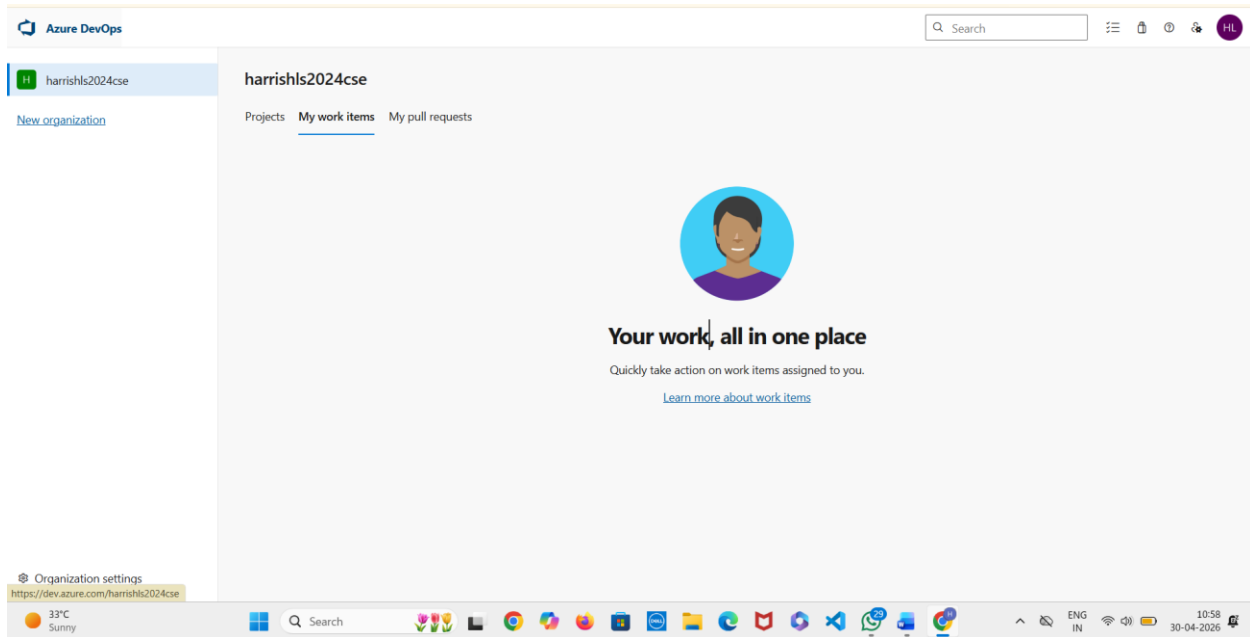
To set up an Azure DevOps project for efficient collaboration and agile work management.

1.Create An Azure Account



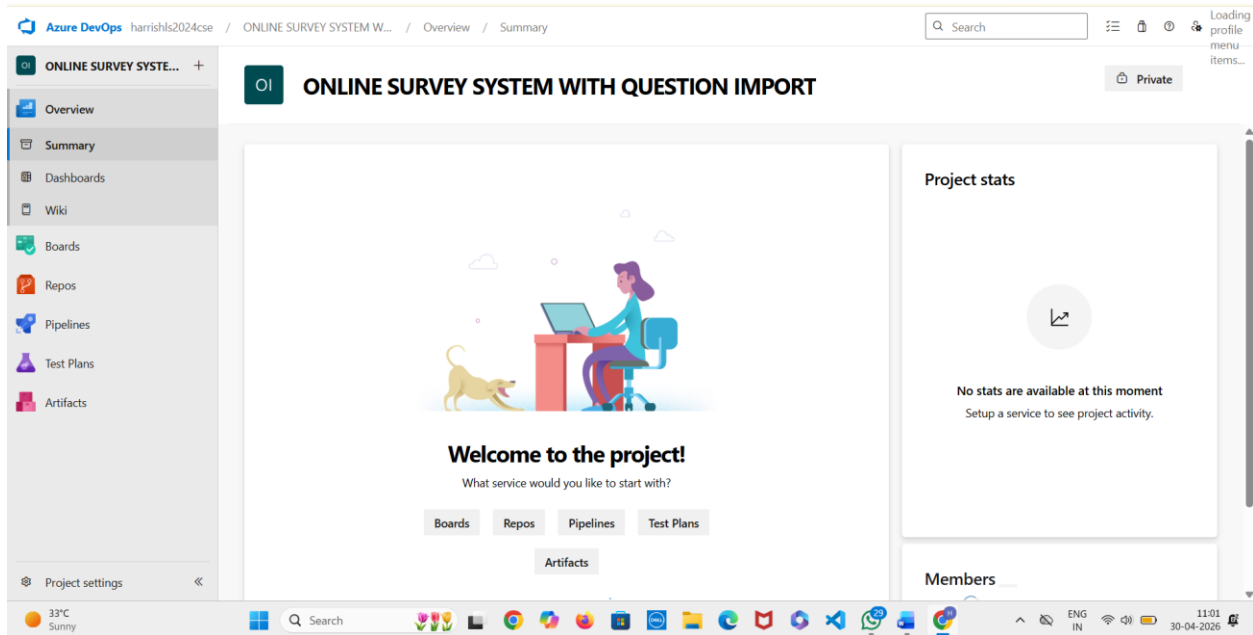
2.Create the First Project in Your Organization

- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:
 - Name:** Choose a name for the project (e.g., **LMS**).
 - Description:** Optionally, add a description to provide more context about the project.
 - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.



3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

4. Project dashboard



5.To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

The screenshot shows the Azure DevOps interface for managing work items. The left-hand navigation menu is visible, with 'Boards' selected. The main area displays the 'Work items' page, which includes a search bar, a '+ New Work Item' button, and a list of work items. The work items are listed in a table with columns for ID, Title, Assigned To, State, Area Path, Tags, and Comments. The work items are numbered 1 through 6, and all are in the 'To Do' state and assigned to 'Unassigned'.

ID	Title	Assigned To	State	Area Path	Tags	Comments
6	As a user, I want to register so that I can create an account.	Unassigned	To Do	ONLINE SURVEY SYSTEM WIT...		
5	User Authentication and Access Control	Unassigned	To Do	ONLINE SURVEY SYSTEM WIT...		
4	Response Analysis and Reporting	Unassigned	To Do	ONLINE SURVEY SYSTEM WIT...		
3	User Response Collection	Unassigned	To Do	ONLINE SURVEY SYSTEM WIT...		
2	Question Management and Import	Unassigned	To Do	ONLINE SURVEY SYSTEM WIT...		
1	Survey Creation and Management	Unassigned	To Do	ONLINE SURVEY SYSTEM WIT...		

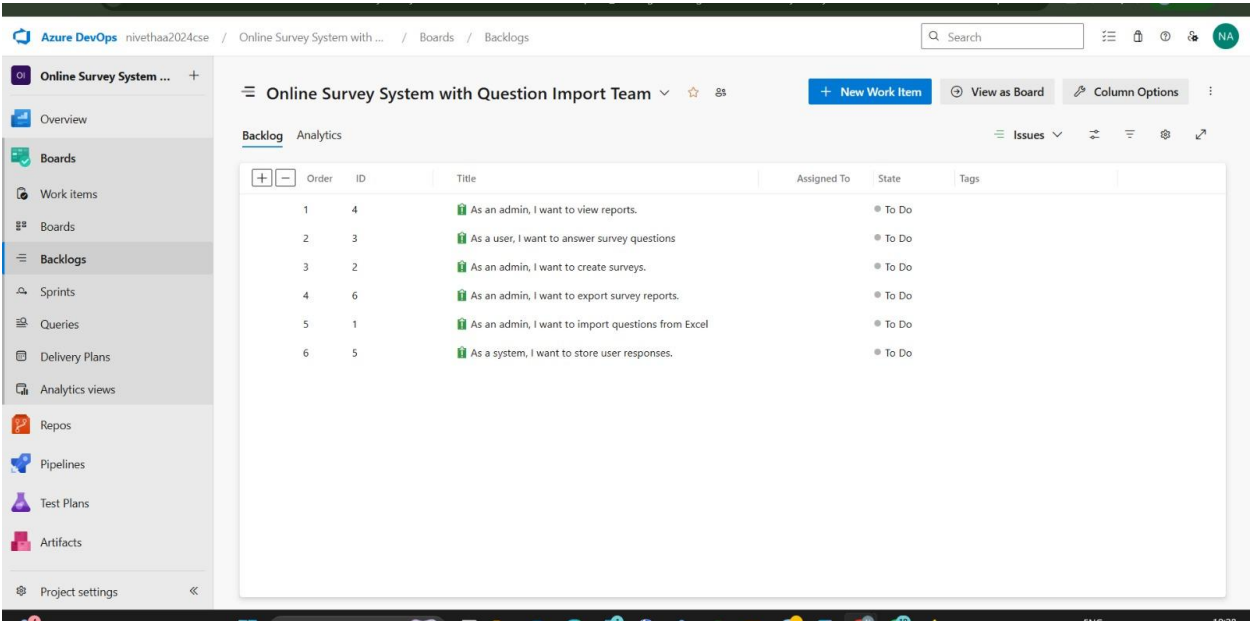
Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

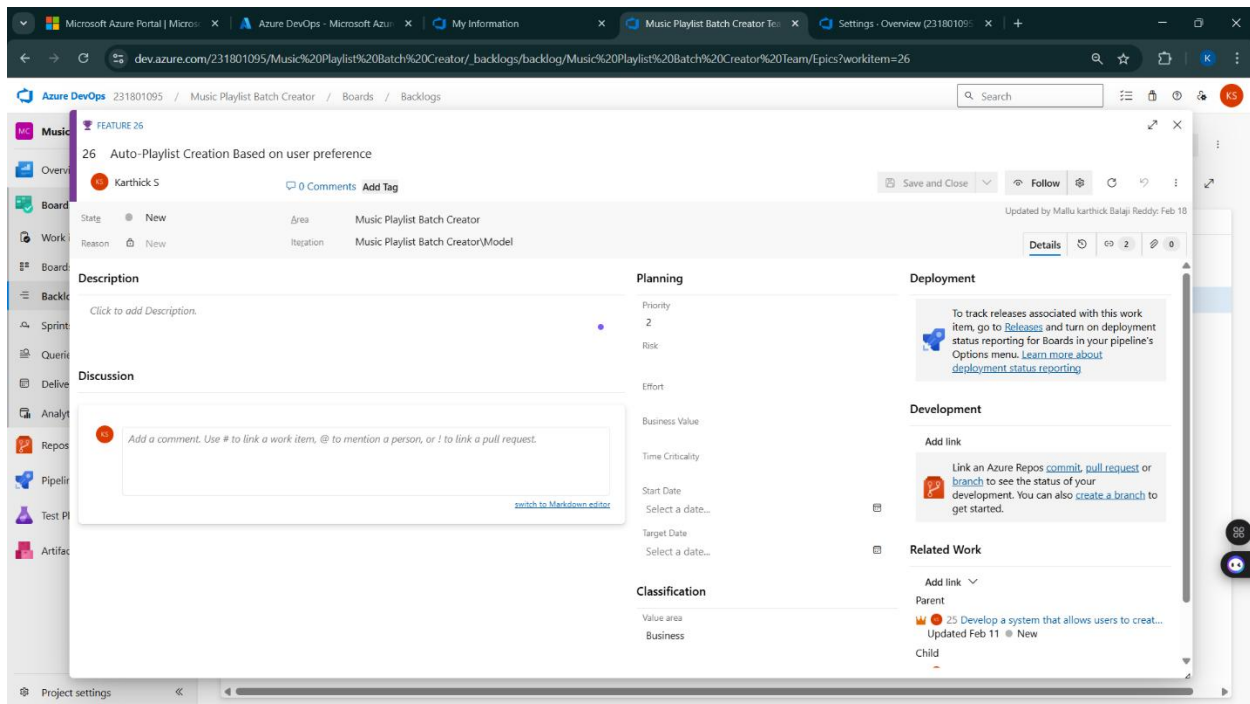
SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:
To learn about how to create epics, user story, features, backlogs for your assigned project.

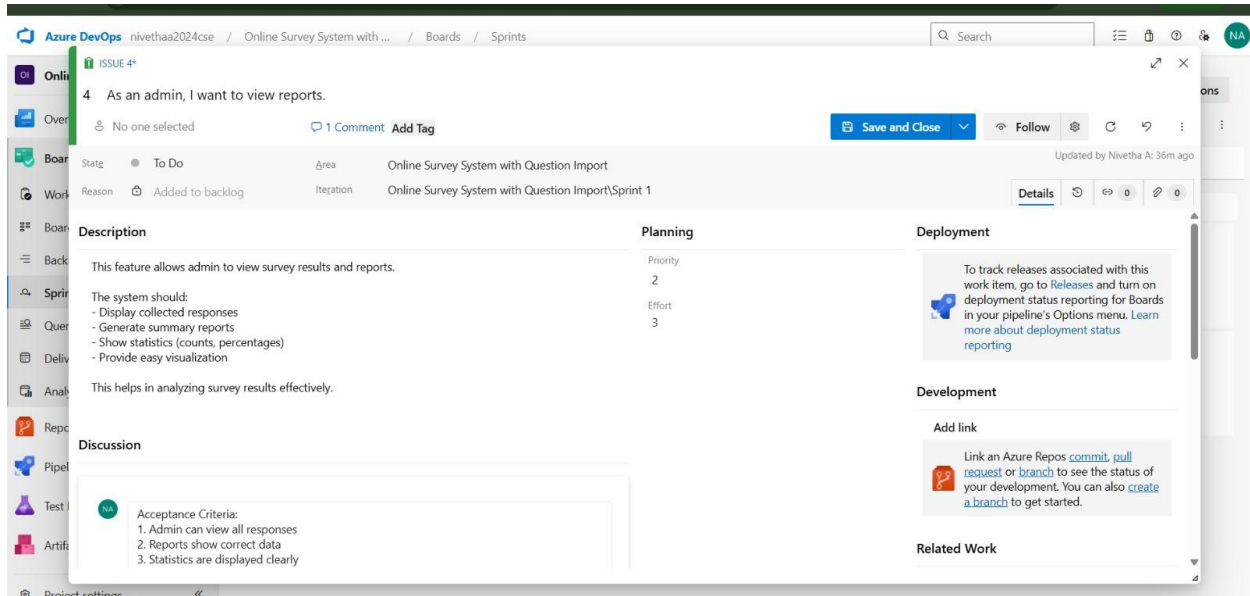
Create Epic, Features, User Stories, Task



1.Fill in Features



3.Fill in User Story Details



Result:

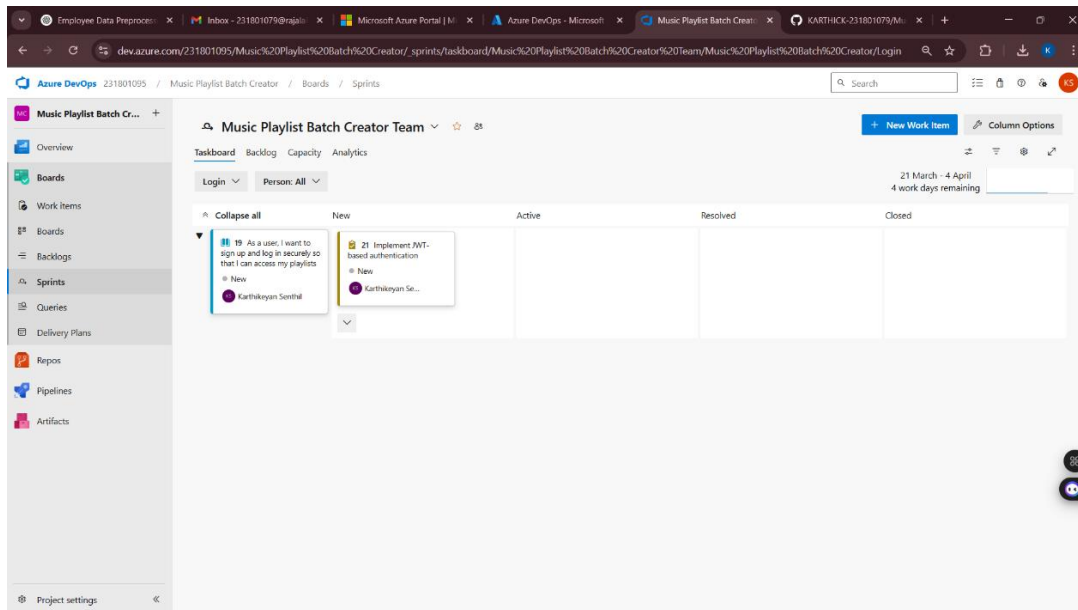
Thus, the creation of epics, features, user story and task has been created successfully.

SPRINT PLANNING

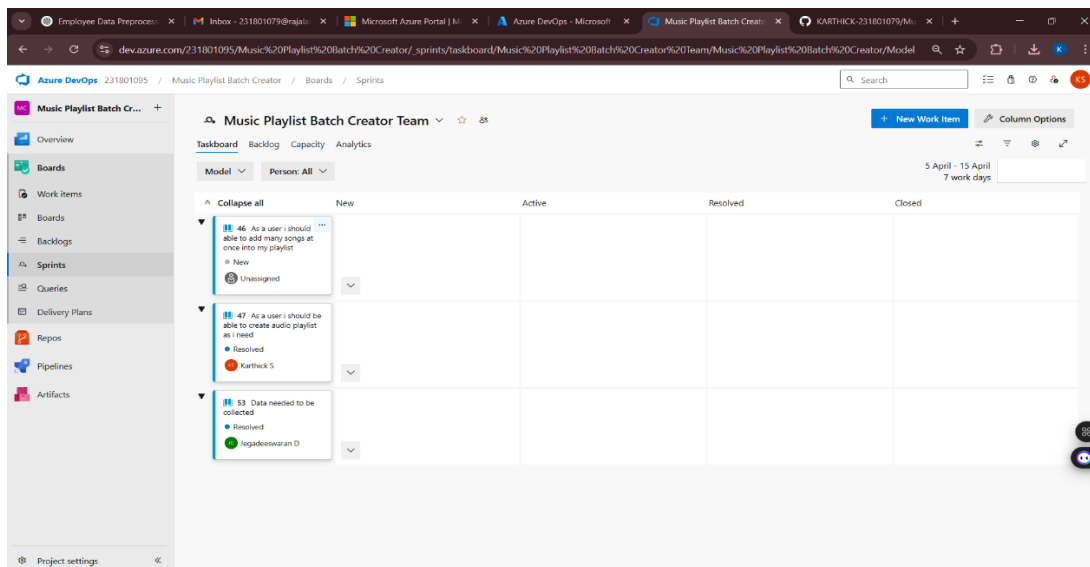
Aim:

To assign user story to specific sprint for the Online survey system Project.

Sprint Planning Sprint 1



Sprint 2



Sprint 3

The screenshot shows the Azure DevOps interface for the 'Music Playlist Batch Creator Team'. The left sidebar contains navigation links: Overview, Boards, Work items, Backlogs, Sprints (selected), Queries, Delivery Plans, Repos, Pipelines, and Artifacts. The main area displays the 'Sprints' board for the period of 15 April to 30 April (12 work days). The board is organized into columns: New, Active, Resolved, and Closed. There are four tasks listed in the 'New' column, all assigned to 'Mallu karthick Balaji R...'. The tasks are:

- 48 As a user, I Can Connect with my music apps (New)
- 20 As a user, I want to link my Spotify account or any music platforms so that I can import my music preferences (New)
- 49 As a user I need to have real time meta data (New)
- 38 As a user I need see my playlist in one place (New)

Sprint 4

The screenshot shows the Azure DevOps interface for the 'Music Playlist Batch Creator Team'. The left sidebar contains navigation links: Overview, Boards, Work items, Backlogs, Sprints (selected), Queries, Delivery Plans, Repos, Pipelines, and Artifacts. The main area displays the 'Sprints' board for the period of 29 April to 30 April (2 work days). The board is organized into columns: New, Active, Resolved, and Closed. There are three tasks listed in the 'New' column, all assigned to 'Karthick S'. The tasks are:

- 23 As a user i can able to customize my playlist (New)
- 50 As a user i should be able to rename ,record and change the playlist (New)
- 51 As a user i should be able to access a friendly and Modern UI (New)

Result:

The Sprints are created for the Music Playlist Batch Creator Project.

POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories – online survey system with question import .

Poker Estimation

The screenshot shows an Azure DevOps work item page for 'USER STORY 47'. The title is '47 As a user i should be able to create audio playlist as i need'. The creator is 'Karthick S'. The status is 'Resolved'. The area is 'Music Playlist Batch Creator'. The reason is 'Code complete and unit t'. The iteration is 'Music Playlist Batch Creator/Model'. The description field is empty with a placeholder 'Click to add Description.'. The acceptance criteria field is empty with a placeholder 'Click to add Acceptance Criteria.'. The discussion field is empty with a placeholder 'Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.'. The planning section shows 'Story Points' as 3, 'Priority' as 2, and 'Risk' as empty. The classification section shows 'Value area' as empty and 'Business' as empty. The deployment section has a link to 'Releases' and a link to 'more about deployment status reporting'. The development section has a link to 'Add link' and a link to 'Link an Azure Repos commit, pull request or branch'. The related work section shows a parent work item '26 Auto-Playlist Creation Based on user preference' updated on Feb 18.

Result:

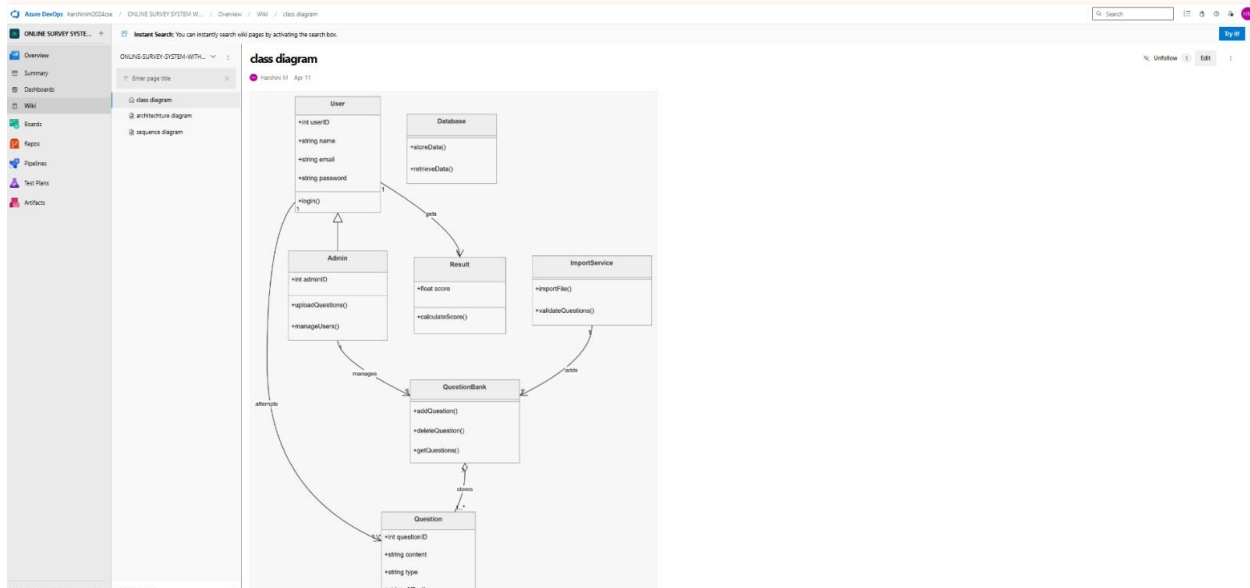
The Estimation/Story Points is created for the project using Poker Estimation.

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

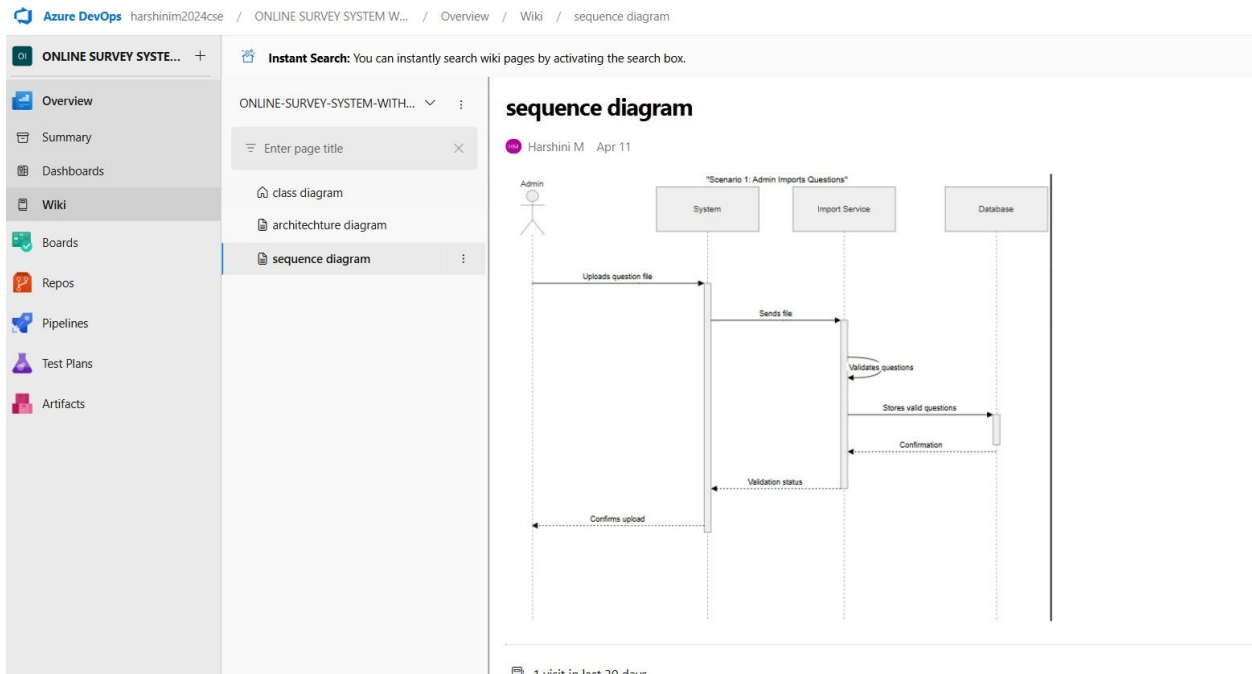
Aim:

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. Class Diagram



6B. Sequence Diagram



Result:

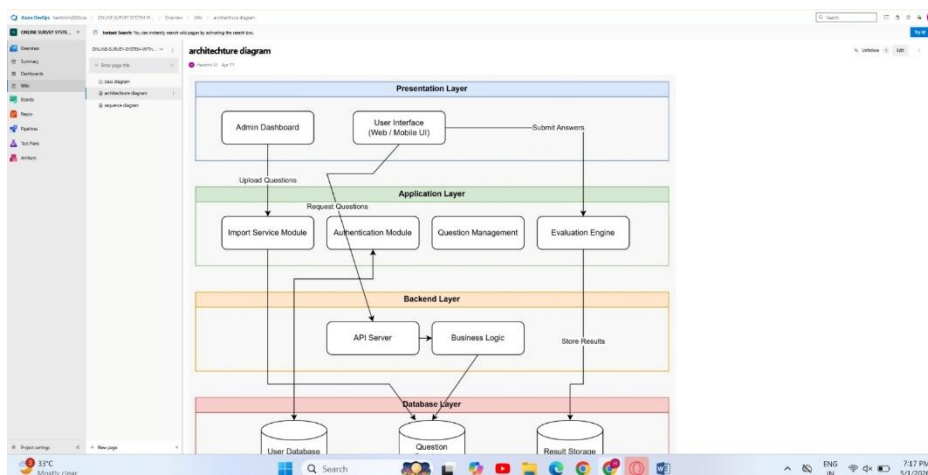
The Class Diagram and Sequence Diagram is designed Successfully for the Music Playlist Batch Creator.

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

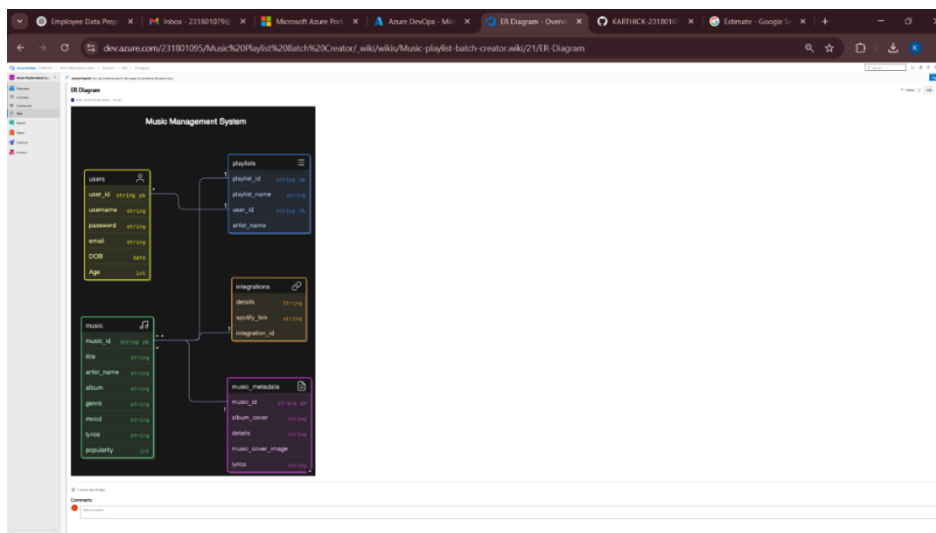
Aim:

To Design an Architectural Diagram and ER Diagram for the given Project.

7A. Architectural Diagram



7B.ER Diagram



Result:

The Architectural Diagram and ER Diagram is designed Successfully for the Music Playlist Batch Creator

	TESTING – TEST PLANS AND TEST CASES
--	--

Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case

Test Case Design Procedure

1. Understand Core Features of the Application

- User Signup & Login
- Viewing and Managing Playlists
- Fetching Real-time Metadata
- Editing playlists (rename, reorder, record)
- Creating smart audio playlists based on categories (mood, genre, artist, etc.)

2. Define User Interactions

- Each test case simulates a real user behaviour (e.g., logging in, renaming a playlist, adding a song).

3. Design Happy Path Test Cases

- Focused on validating that all features function as expected under normal conditions.
- Example: User logs in successfully, adds item to playlist, or creates a category-based playlist.

4. Design Error Path Test Cases

- Simulate negative or unexpected scenarios to test robustness and error handling.
- Example: Login fails with invalid credentials, save fails when offline, no recommendations found.

5. Break Down Steps and Expected Results

- Each test case contains step-by-step actions and a corresponding expected outcome.
- Ensures clarity for both testers and automation scripts.

6. Use Clear Naming and IDs

- Test cases are named clearly (e.g., TC01 – Successful Login, TC10 – Save Playlist Fails).
- Helps in quick identification and linking to user stories or features.

7. Separate Test Suites

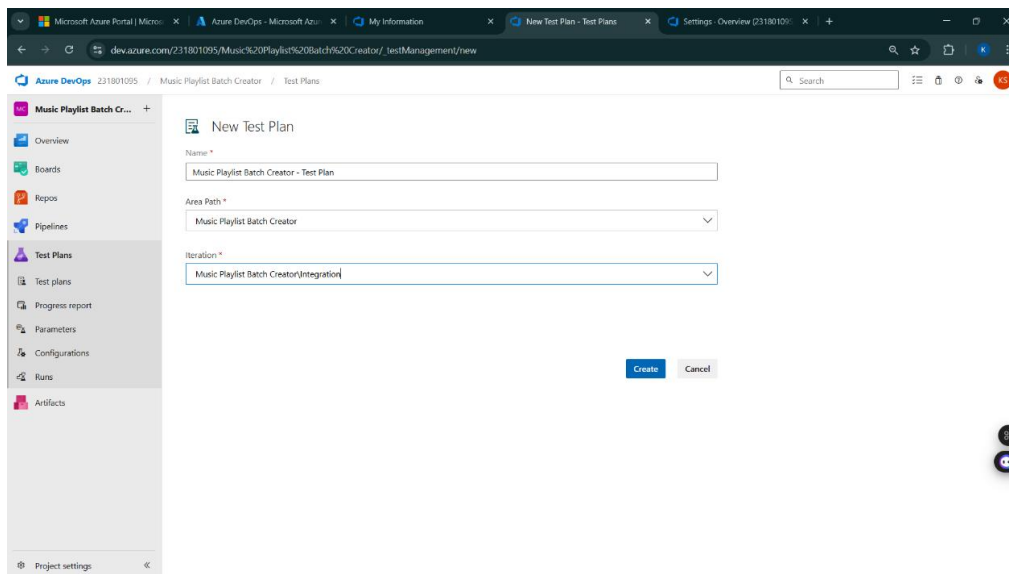
- Grouped test cases based on functionality (e.g., Login, Playlist Editing, Recommendation System).

- Improves organization and test execution flow in Azure DevOps.

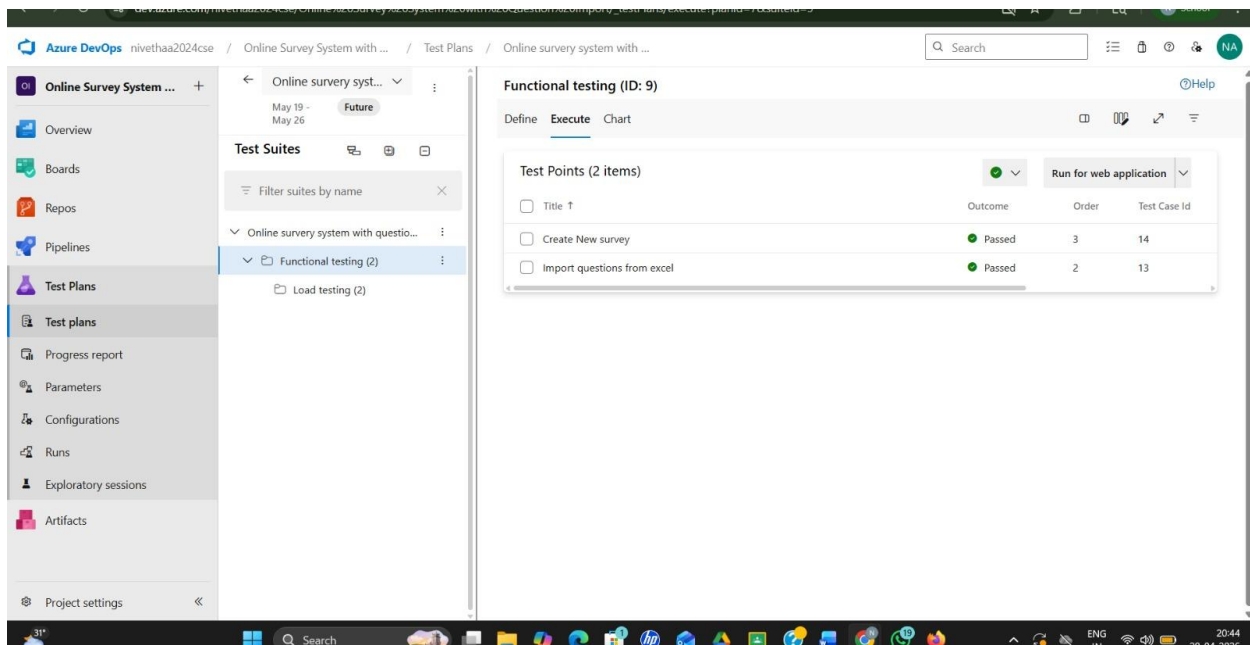
8. Prioritize and Review

- Critical user actions are marked high-priority.
- Reviewed for completeness and traceability against feature requirements.

1.New test plan



2.Test suite



Online Survey System – Test Plans

USER STORIES

- **As an admin, I want to import questions so that I can create surveys quickly (ID: 101).**
- **As an admin, I want to create a survey so that I can collect responses (ID: 102).**
- **As a user, I want to take a survey so that I can submit my responses (ID: 103).**
- **As an admin, I want to view survey results so that I can analyze responses (ID: 104).**
- **As a user, I want to login securely so that I can access surveys (ID: 105).**

Test Suites

Test Suit: TS01 - User Login (ID: 201)

1. TC01 – Successful Login

o Action:

- **Go to the Login page.**
- **Enter valid username and password.**
- **Click "Login".**

o Expected Results:

- **Login form is displayed.**
- **Fields accept valid input.**
- **User is redirected to dashboard.**

o Type: Happy Path

2. TC02 – Login Failure

o Action:

- **Go to the Login page.**
- **Enter valid username and incorrect password.**
- **Click "Login".**

o Expected Results:

- **Input is accepted.**
- **Error message "Invalid credentials" is displayed.**

o Type: Error Path

Test Suit: TS02 - Question Import (ID: 202)

1. TC03 – Import Questions Successfully

o Action:

- Login as admin.
- Navigate to "Import Questions".
- Upload valid CSV file.
- Click "Import".

o Expected Results:

- File is uploaded successfully.
- Questions are added to database.

o Type: Happy Path

2. TC04 – Import Failure Due to Invalid File

o Action:

- Login as admin.
- Navigate to "Import Questions".
- Upload invalid file format.
- Click "Import".

o Expected Results:

- Error message "Invalid file format" is displayed.

o Type: Error Path

Test Suit: TS03 - Create Survey (ID: 203)

1. TC05 – Create Survey Successfully

o Action:

- Login as admin.
- Click "Create Survey".
- Enter survey details.
- Select questions.
- Click "Save".

o Expected Results:

- Survey is created successfully.

o Type: Happy Path

2. TC06 – Create Survey Failure

o Action:

- Login as admin.
- Click "Create Survey".
- Leave required fields empty.
- Click "Save".

o Expected Results:

- Error message "Fields cannot be empty" is displayed.

o Type: Error Path

Test Suit: TS04 - Take Survey (ID: 204)

1. TC07 – Submit Survey Successfully

o Action:

- Login as user.
- Open available survey.
- Answer all questions.
- Click "Submit".

o Expected Results:

- Survey responses are submitted successfully.

o Type: Happy Path

2. TC08 – Submit Survey Failure

o Action:

- Login as user.
- Open survey.
- Skip mandatory questions.
- Click "Submit".

o Expected Results:

- Error message "Please answer all required questions" is displayed.

o Type: Error Path

Test Suit: TS05 - View Results (ID: 205)

1. TC09 – View Results Successfully

o Action:

- Login as admin.
- Navigate to "Survey Results".

o **Expected Results:**

- **Survey responses are displayed correctly.**

o **Type: Happy Path**

2. TC10 – View Results Failure

o **Action:**

- **Login as admin.**
- **Navigate to "Survey Results" without data.**

o **Expected Results:**

- **Message "No responses available" is displayed.**

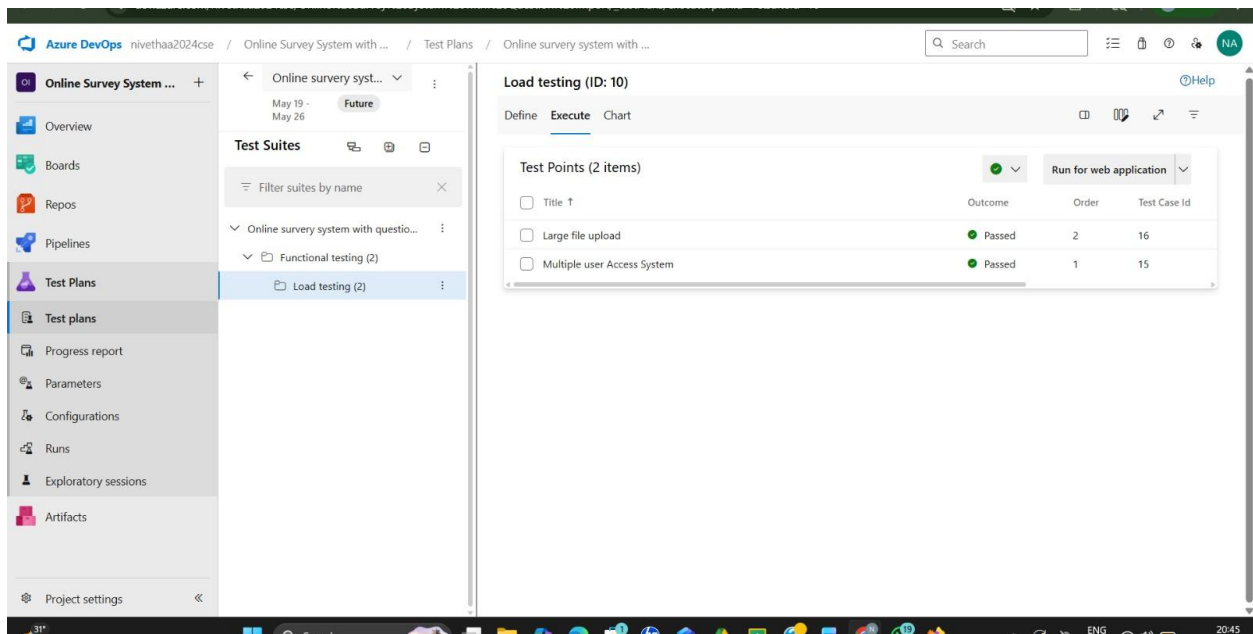
o **Type: Error Path**

Test Cases

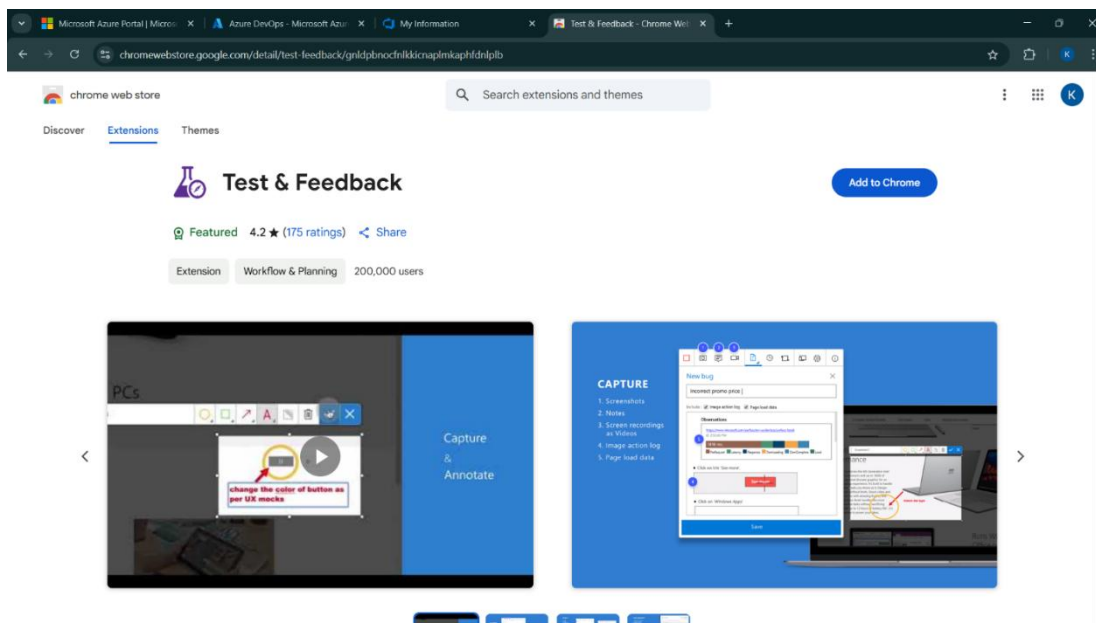
The screenshot displays the Azure DevOps Test Plans interface. The left sidebar shows the navigation menu with 'Test Plans' selected. The main area shows a test suite named 'Functional testing (2)' under the 'Online survey system with question...' project. The test suite contains two test points:

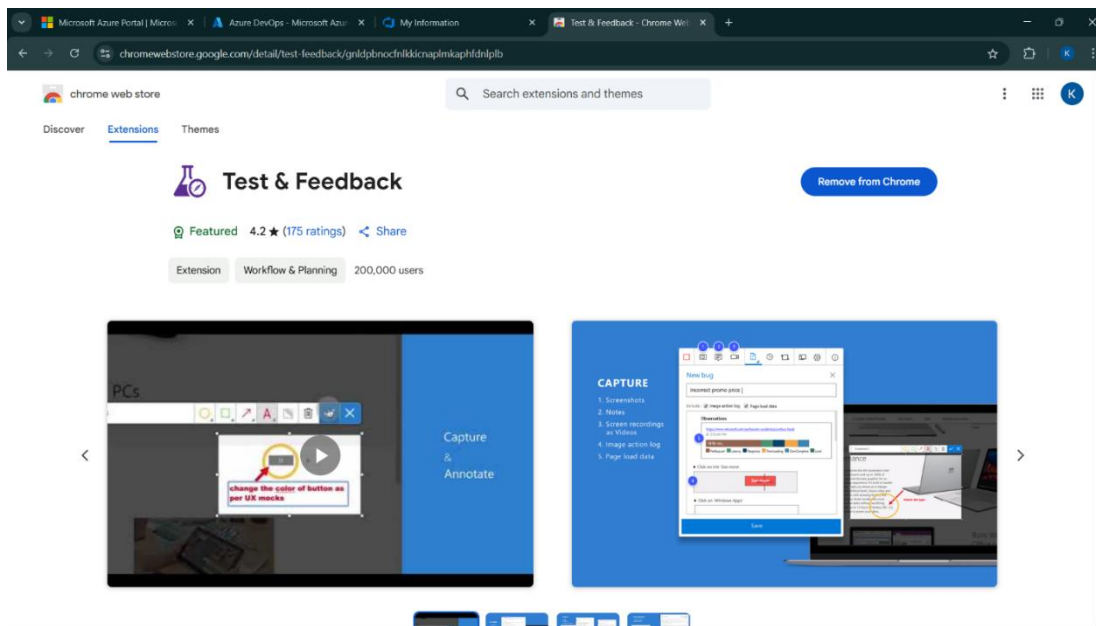
Title	Outcome	Order	Test Case Id
Create New survey	Passed	3	14
Import questions from excel	Passed	2	13

The interface also shows a 'Run for web application' button and a 'Test Points (2 items)' section. The bottom status bar indicates the system is running on Windows 10, with the date and time set to 20:44 on 29-04-2026.

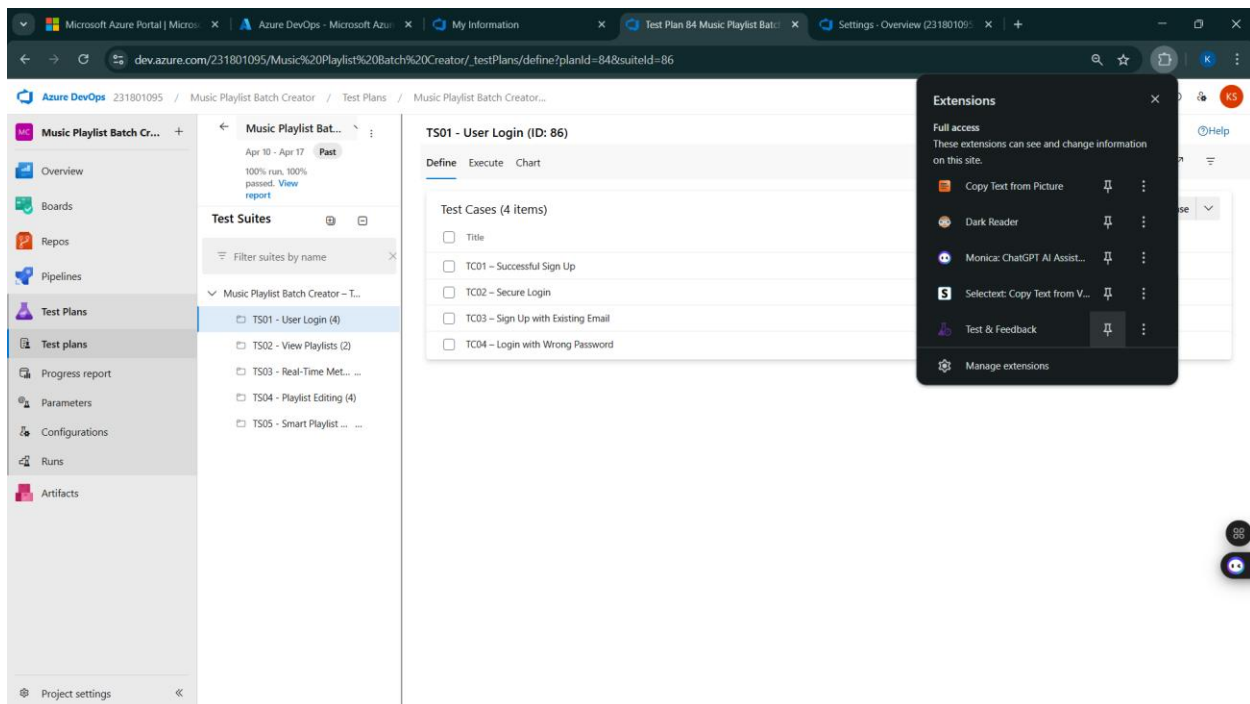


4.Installation of test





Test and feedback
Showing it as an extension



5. Running the test cases

The screenshot shows the Azure DevOps interface with the 'Test Plans' section selected. A modal titled 'Test point ID 5 execution history' is open, displaying the following information:

- Test Case: 15 Multiple user Access System
- Test Suite: 10 Load testing
- Current Tester: Nivetha A
- Configuration: Windows 10

Outcome	Test run	Run by	Time completed
Passed	4	Nivetha A	Just now

Currently showing a test point execution history. View all history for a full test case execution history. [View all history](#)

8. Test case results

The screenshot shows the Azure DevOps interface with the 'Test Plans' section selected. A modal titled 'TC05 - View Playlist Page' is open, displaying the following information:

- Test Case: TC05 - View Playlist Page
- Test Suite: TS02 - View Playlists (ID: 87)
- Configuration: Windows 10
- Run by: Karthick S
- Tester: Mallu karthick Balaji ... Music

Outcome	TimeStamp	Configuration	Run by	Tester	Test P
Passed	4m ago	Windows 10	Karthick S	Mallu karthick Balaji ... Music	
Passed	12m ago	Windows 10	Karthick S	Mallu karthick Balaji ... Music	
Not Applicable	12m ago	Windows 10	Karthick S	Mallu karthick Balaji ... Music	
Passed	14m ago	Windows 10	Karthick S	Mallu karthick Balaji ... Music	
Passed	Tuesday	Windows 10	Karthikeyan Senthil	Mallu karthick Balaji ... Music	
Passed	Saturday	Windows 10	Mallu karthick Balaji ... Music	Mallu karthick Balaji ... Music	
Failed	Saturday	Windows 10	Mallu karthick Balaji ... Music	Mallu karthick Balaji ... Music	
Passed	Apr 11	Windows 10	Karthick S	Mallu karthick Balaji ... Music	
Passed	Apr 11	Windows 10	Karthick S	Mallu karthick Balaji ... Music	

[Open execution history for current test point](#)

9. Test report summary

The screenshot shows the Azure DevOps interface for a bug report titled "BG 01 - Countries Drop down Not Available on the page". The bug is assigned to "rajesh prabhu" and is in the "New" state. A dropdown menu for "Repro Step" is open, showing options: "New", "Active", "Resolved", and "Closed". The "New" option is selected. The bug description states: "Validated list of countries displayed on the". The "Repro Steps" table shows three steps:

Step no.	Result	Title
1.	Passed	Open Chrome Browser
2.	Passed	Paste the URL (https://127.0.0.1/new.html)
3.	Failed	Expected Result: App should load with new articles. Actual Result: Screen should show a screen with list of countries. Action: Produce Report

The right sidebar contains sections for "Planning", "Deployment", "Development", and "Related Work".

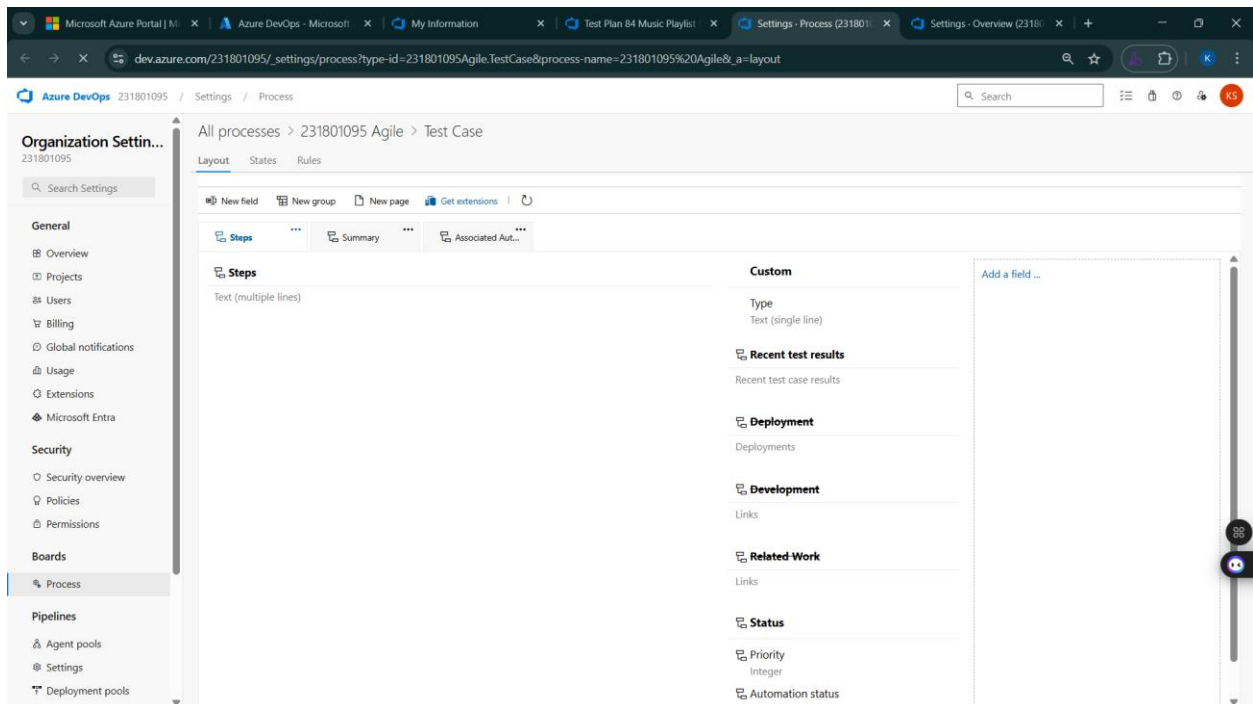
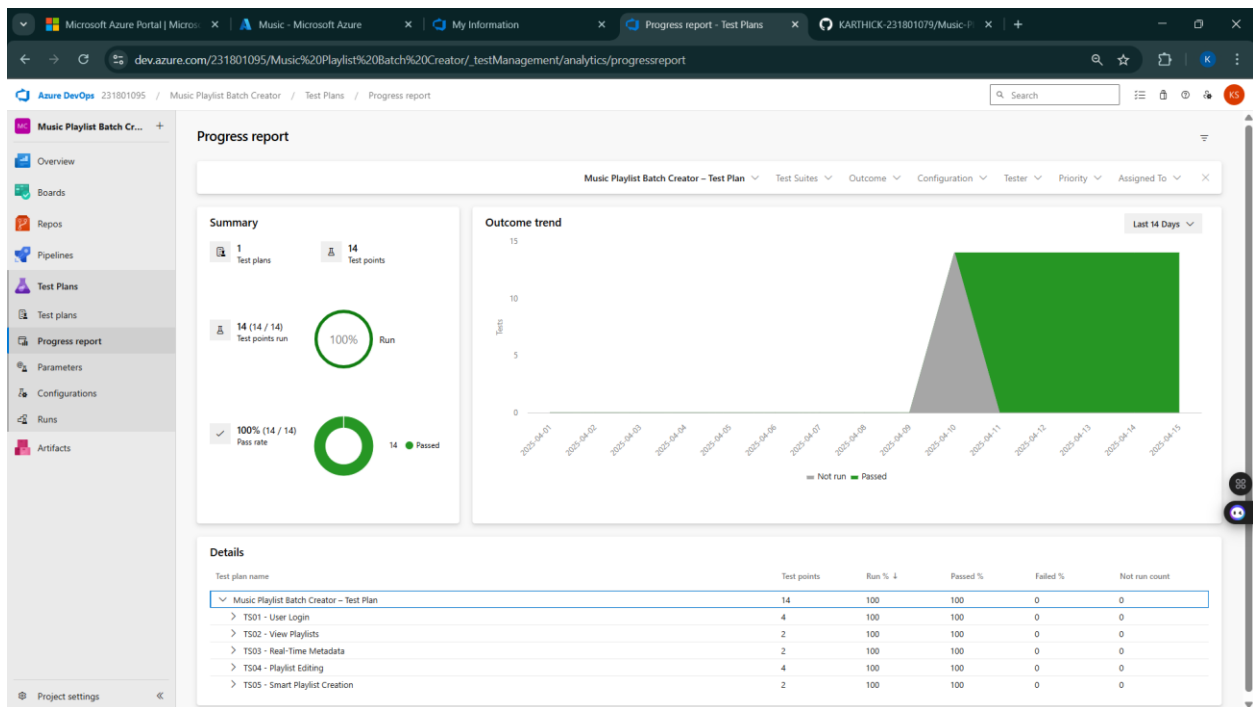
- Assigning bug to the developer and changing state

The screenshot shows the Azure DevOps interface for a bug report titled "TB01 - Playlist loading spinner keeps spinning indefinitely on poor network". The bug is assigned to "Karthick S" and is in the "New" state. The "Repro Steps" table shows two steps:

Step no.	Result	Title
1.	Failed	Disconnect from internet
2.	Failed	Comments: Page Not loading. Navigate to "My Playlists". Expected Result: Error message "Unable to load playlists" is shown.

The right sidebar contains sections for "Planning", "Deployment", "Development", and "Related Work".

10. Progress report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

	LOAD TESTING AND PIPELINES
--	-------------------------------------

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource
 - o Go to *Create a resource* → Search for “Azure Load Testing”.
 - o Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
 - o *Subscription*: Choose your Azure subscription.
 - o *Resource Group*: Create new or select an existing one.
 - o *Name*: Provide a unique name (no special characters).
 - o *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

Steps to Create and Run a Load Test:

Once your resource is ready:

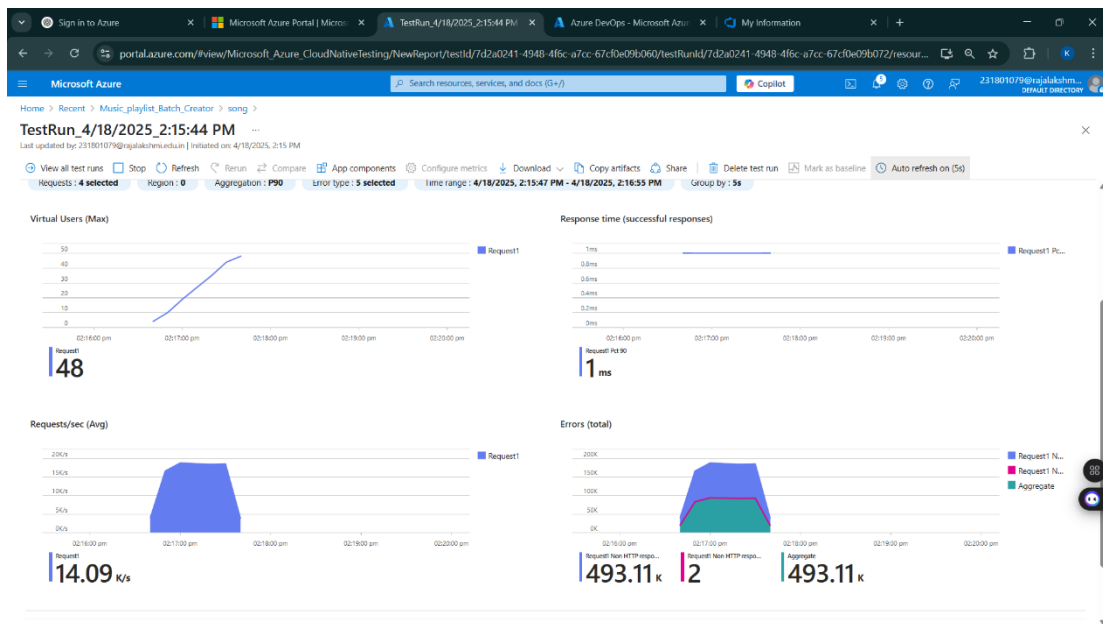
1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab
 - o *Test Name*: Provide a unique name.
 - o *Description*: (Optional) Add test purpose.
 - o *Run After Creation*: Keep checked.
3. Load Settings
 - o *Test URL*: Enter the target endpoint (e.g., <https://yourapi.com/products>).

- Click Review + Create → Create to start the test.

Load Testing

The screenshot shows the Azure DevOps Test Plans interface. On the left, the 'Test Plans' section is expanded, showing 'Test plans', 'Progress report', 'Parameters', 'Configurations', 'Runs', 'Exploratory sessions', and 'Artifacts'. The main area displays a 'Load testing (ID: 10)' suite. The 'Test Points (2 items)' table shows the following data:

Title	Outcome	Order	Test Case Id
Large file upload	Passed	2	16
Multiple user Access System	Passed	1	15



Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted Flask-based music recommendation project with Azure DevOps. The pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
 - In Azure DevOps, create a new project.
 - Create a pipeline and select GitHub as the source.
 - Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
 - In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - Add the following basic pipeline configuration for Python and Flask:

yml Code

trigger:

```
- main # Runs pipeline when code is pushed to main branch
```

pool:

```
vmImage: ubuntu-latest # Uses Ubuntu agent
```

steps:

```
# Step 1: Checkout code
```

```
- checkout: self
```

```
# Step 2: Set up Python environment
```

```
- task: UsePythonVersion@0
```

```
inputs:
```

```
versionSpec: '3.x'
```

```
displayName: "Set up Python"
```

```
# Step 3: Install dependencies
```

```
- script: |
```

```
python -m pip install --upgrade pip
pip install -r requirements.txt
displayName: "Install dependencies"
```

Step 4: Run basic validation script

- script: |

```
python -c "print('🎉 Online Survey System with Question Import is running successfully!')"
displayName: "Run validation script"
```

Step 5: (Optional) Run tests

- script: |

```
echo "Running test cases for survey system..."
displayName: "Run tests"
```

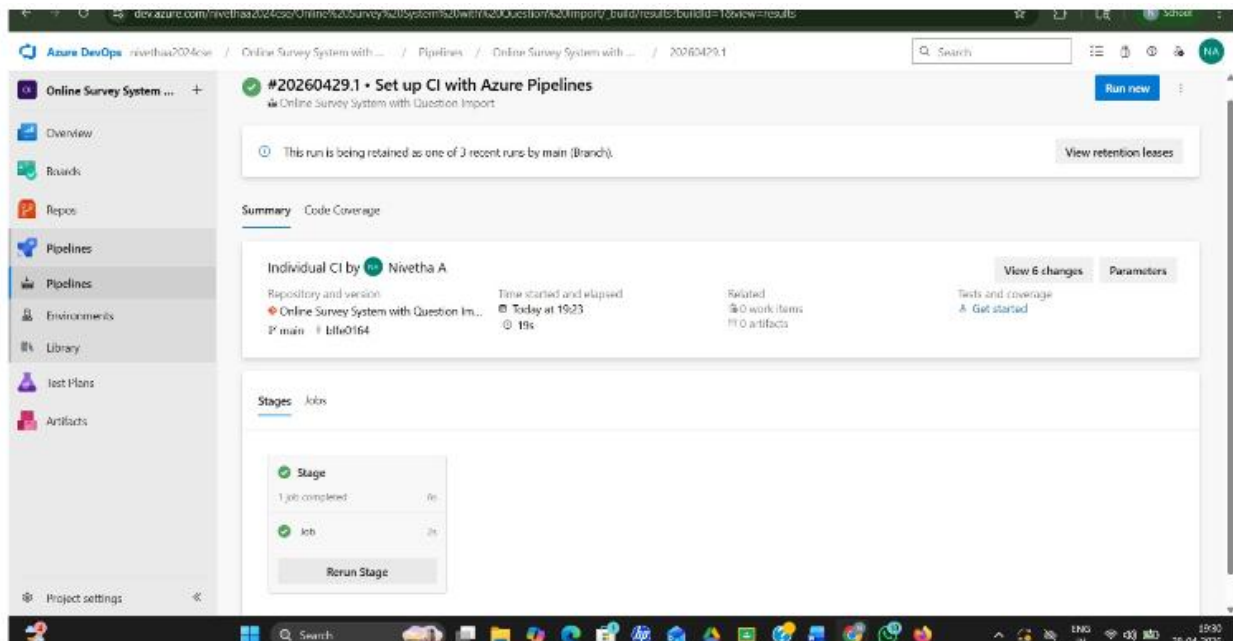
3. Pipeline Tasks Include:

- Setting up the Python environment using the UsePythonVersion task.
- Installing project dependencies from project/requirements.txt. Make sure the path to requirements.txt is correct (it is located under the project folder).
- Running a simple Python script to verify that Python is set up correctly and the pipeline works.

4. Run and Monitor Pipeline:

- Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
- Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline



Result:

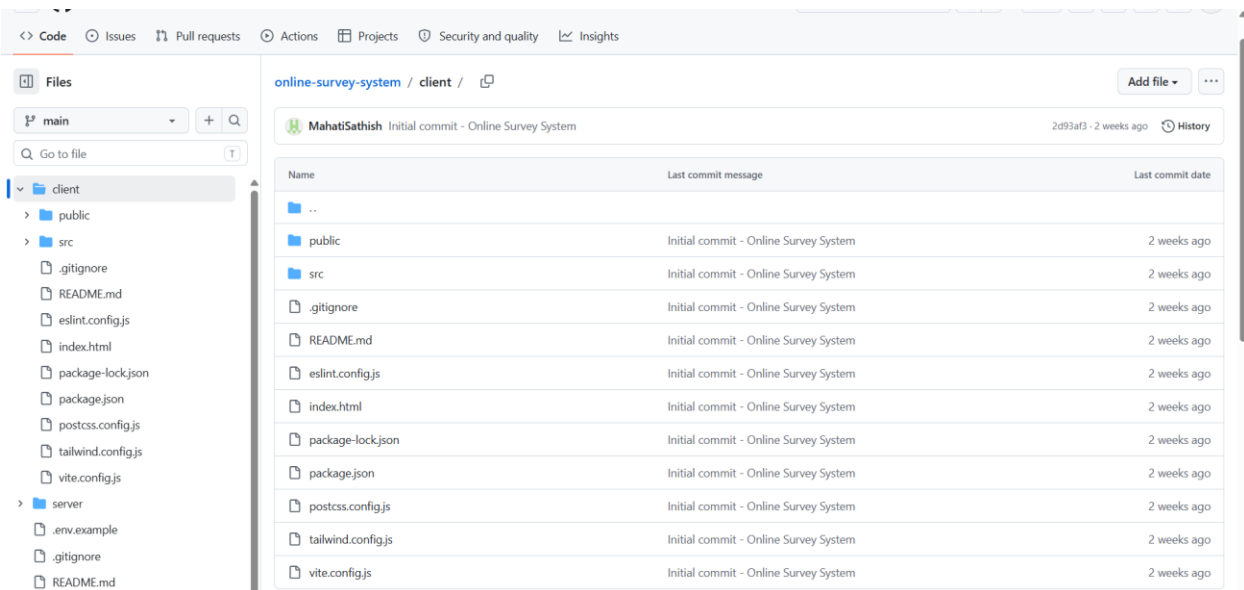
Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Online survey system project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.